



Response to Request for Information on the Development of an Artificial Intelligence Action Plan

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The University of Rochester is among America's leading research universities and has a long history in computer science, data science, machine learning and artificial intelligence research. The University of Rochester Medical Center is the largest integrated health system in Upstate New York home to eight hospitals, 7 nursing homes, the National Cancer Institute's latest designated cancer center, and the region's largest oral health safety net provider.

Artificial intelligence (AI) is a transformative and increasingly pervasive technology that is changing our lives in fundamental and profound ways. As part of the University's 2030 strategic plan, Boundless Possibility, Rochester faculty, staff, and students are integrating and leveraging AI in their work to enhance how we learn, discover, heal, and create. The University has significant expertise and is making significant investments in this area, including:

- Rochester offers a BS, BA, MS, and PhD in computer science and a BA, BS, and MS in data science through the Hajim School of Engineering and Applied Sciences. And Simon Business School offers an online MS in business analytics and applied AI. Additional degree and certification programs are available and forthcoming.
- The Center of Excellence (CoE) in Data Science and Artificial Intelligence. The CoE is dedicated to supporting businesses in New York through the application of data science methods and tools that solve challenges and deliver critical insights. To date, that's resulted in an economic impact of more than \$140 million.
- The Goergen Institute for Data Science and Artificial Intelligence serves as an interdisciplinary hub, creating new opportunities for Rochester faculty and students to collaborate and stay on the cutting edge of AI research.
- Rochester recently joined Empire AI - a New York statewide consortium of leading public and private academic institutions, who are collaborating to facilitate groundbreaking AI

research and innovative technological development. As the consortium's only integrated health system, we are driving the advancement of AI-powered health care.

- Rochester recently acquired a tier 3 data center, increasing our capacity for machine learning and AI research. In addition, the University's Laboratory for Laser Energetics acquired (with Federal and State support) Conesus, a new super-computer which quadrupled available computing power while being among the world's most energy-efficient computer systems. Conesus is dedicated to national security research in the areas of inertial confinement fusion and high-energy-density physics research.
- The University of Rochester Medical Center has created its own foundational AI architecture platform HyperionAI offering secure, on-premise access to widely used generative AI models without the need for additional hardware or a complicated software environment, all with the confidence of maintaining healthcare care data privacy and security. HyperionAI also powers our Generative AI Healthcare Data Chatbot Chat HAI. ChatHAI functions similarly to genAI platforms like ChatGPT and can handle data from a wide range of sources while also providing full data privacy and security through features such as automatic PHI-scrubbing to ensure that critical, identifiable, healthcare information is safe.

Recommendations

The University appreciates the opportunity to comment on the development of an Artificial Intelligence (AI) Action Plan and offers the following recommendations on priority actions that should be included in the Plan to enhance America's global AI dominance to promote human flourishing, economic competitiveness, and national security:

- 1) *AI Research Infrastructure* - Support networking of computing, data, and AI/ML capabilities across research universities to complement approaches taken by national labs, including NAIRR.
- 2) *AI Education and Workforce Development* - Grow targeted AI STEM education and workforce training programs across federal agencies leveraging existing programs and provide opportunities for AI developers and AI users.
- 3) *AI in Medicine to Promote Human Flourishing* - Turbocharge AI for biomedical and health applications by increasing support for existing program and making targeted investments in promising areas like intelligent imaging.
- 4) *AI for Fusion to Promote National and Energy Security* - AI for fusion energy program to accelerate research and reduce barriers to fusion energy commercialization.
- 5) *AI Research* – Maintain support for core research, especially at NSF, and support for a broader DOE AI initiative to address science, energy, and national security challenges.

AI Research Infrastructure

Rochester recommends the Plan support the availability of right-sized shared AI resources for the academic community. Given the rapidly increasing costs of AI optimized computing hardware, along with the computing resource requirements of leading-edge AI tools, many universities, including Rochester face constraints that slow progress in both research and education. For instance, available AI computing resources may be underpowered for researchers seeking to address AI development questions at the forefront of the field, while being overpowered when time is assigned for student homework or class projects. Rochester supports the NSF National Artificial Intelligence Research Resource (NAIRR) and is participating in the pilot. The Plan should prioritize implementation of the full-scale NAIRR while guiding development to ensure NAIRR resources are available at a variety of scales needed for research and education.

The University recently acquired an existing and under-utilized tier 3 data center in the City of Rochester and is in the process of refurbishing and modernizing the facility. The immediate focus is meeting the data processing needs of the University and Medical Center, however, we anticipate the final facility will have significant excess capacity which could be made available to outside users. The Plan should consider the existence of currently under-utilized purpose-built AI related infrastructure and how to best network these and create a shared user mode to leverage existing resources to best meet rapidly growing demand for data centers and other AI-related infrastructure as quickly as possible.

AI Education and Workforce Development

Rochester recommends the Plan call for expanded support for existing training opportunities for graduate students, post-doctoral researchers and early career faculty, including, NSF Graduate Research Fellowship Program, DOE Computational Science Graduate Fellowship Program and the Early Career Research Program in fields critical for AI development. In addition to support for these existing programs, the Plan should ensure that training opportunities are expanded to explicitly include fields that will be users of AI as well. As an example, Rochester's Warner School of Education and Eastman School of music conducted an NSF funded project that studied the emerging occupation of "artist-technologist". As experts in both arts and technology who see no barriers between these two fields, artist-technologists are in a unique position to design previously inconceivable media products with relevance and applications, not only for arts and entertainment but for many other fields as well, including education, health care, marketing, and new product development in most industries. Similarly, Rochester's Simon School of Business has launched a comprehensive AI initiative, integrating generative AI across programs through training, curriculum, and workshops and Full-Time MS in Artificial Intelligence in Business. This initiative will help students to master the cutting-edge AI technologies transforming industries, preparing them to better leverage AI for business value creation.

Because AI tools will impact everyone and be used in ways we currently cannot anticipate the Plan should support all students in being better prepared to incorporate AI into their future careers.

AI in Medicine to Promote Human Flourishing

AI has tremendous potential to advance biomedical research, improve clinical care, utilize existing clinical and research data to provide novel insights, and improve medical education. For more than a decade, the University of Rochester Medical Center (URMC) has been a leader in incorporating AI tools into health research and clinical care, through tools such as HyperionAI and ChatHAI. The UR Health Lab engages in institution-wide, high-impact projects and utilizes data and technology to innovate healthcare solutions to find the best solutions to complex problems that lead to better healthcare outcomes. The Goergen Institute for Data Science and Artificial Intelligence is working with our health system to build non-generative AI models for ultrasound and other imaging uses. We have predictive models that help to screen patient lab work and flag results that are at high risk for sepsis and other conditions with significant mortality rates and are beginning to roll out ambient documentation tools to assist providers with taking notes during patient appointments.

Rochester recommends the Plan call for increased support for AI research across NIH and at ARPA-H. This should include core research programs across the various institutes. Rochester supports programs, such as Bridge to Artificial Intelligence (Bridge2AI), that will provide data sets and best practices for machine learning analysis for use by the broader community. AI efforts in health should also provide support for education and understanding the ethical and societal implications of advanced AI so that all may benefit from these advances.

Medical imaging is an area that is critical to patient outcomes and can substantially benefit from incorporation of AI to provide more accurate diagnostics, more rapid diagnostics, personalized treatment planning, simplified workflows, improved quality assurance, the ability to bring radiological capabilities to remote and underserved areas. The Plan should consider creation of one or more centers focused on intelligent imaging, the application of artificial intelligence for image generation, analysis, interpretation, and utilization, in healthcare to best scale on-going efforts across the country. Efforts at such a center should include innovation, translation to practice, and education and workforce development.

AI for Fusion to Promote National and Energy Security

Rochester recommends a dedicated program to expand the incorporation of AI into fusion research and development. Such a program should be jointly managed by DOE Office of

Science and the National Nuclear Security Administration (NNSA). Rochester's Laboratory for Laser Energetics (LLE) is one of three principal inertial confinement fusion (ICF) program sites funded by NNSA to conduct important stockpile stewardship research. LLE also leads the DOE Fusion Energy Sciences funded Inertial Fusion Energy-Consortium on LPI Research (IFE-COLoR) hub and leads the Inertial Fusion Energy Science and Technology Accelerated Research (IFE-STAR) ecosystem that brings together academia, national laboratories, and the private sector to develop a clean, safe, and virtually limitless energy source, built on US leadership in inertial fusion.

LLE, working with national lab, academic, and industrial partners has made significant advances in utilizing AI and machine learning to increase fusion yield, improve modeling performance, develop better targets, and accelerate data analysis. LLE foresees future uses of AI in improving laser (or other fusion driver) and facility operation. LLE believes that future fusion energy power plants, as well as future high energy density research facilities will necessarily operate at high shot rates only achievable through substantial integration of AI and computer control into the operating environment. Rochester believes an AI for fusion program should support development of AI tools and methods for fusion research and facility operation, as well as training opportunities for researchers and operators.

AI Research

Rochester recommends the Plan increase investments in existing AI programs, including those at NSF, including basic science funded by core programs, the Major Research Instrumentation Program to better expand access to hardware for AI research, and fund additional National Artificial Intelligence Research Institutes to advance multidisciplinary, multi-stakeholder research on a larger scale.

Rochester also recommends the Plan consider a dedicated DOE AI Initiative, aligned with a broader whole-of-government approach to AI research and development, to fully leverage DOE's unique infrastructure and expertise to meet the nation's science, energy, and national security challenges. Rochester supports the related comments from the Energy Sciences Coalition.

Thank you for the opportunity to comment on the development of an Artificial Intelligence (AI) Action Plan and for considering these recommendations.

Sincerely,

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